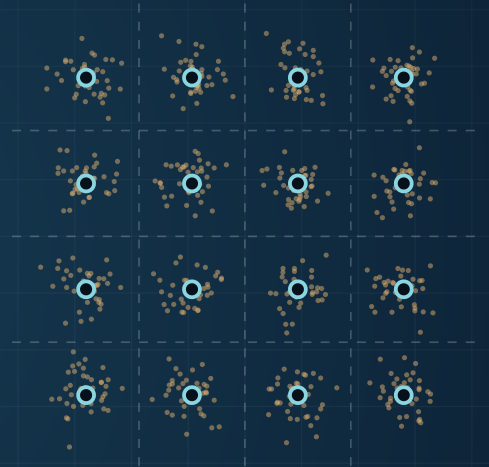
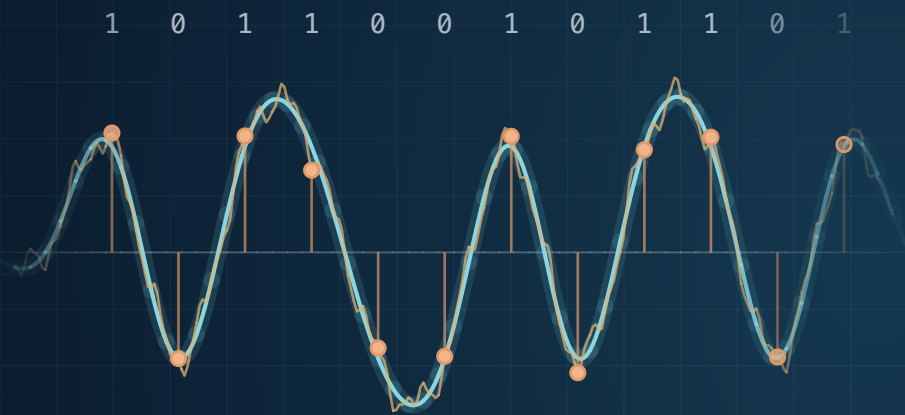




SAMPLING · DETECTION · MODULATION · CODING

Digital Communications

Student Workbook



Contents

Every question in the course, with no answer and no solution. Work each one on the page, then check it against the artifact or against the instructor edition.

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How to use it

The questions are in the order the course meets them, and each is labelled with what it asks for. Only the statement and its lettered parts are printed; the reasoning stays for you to supply. The question numbers are shared with every other edition, so D5-04 is the same question in the artifact, in this workbook and in the instructor solutions.

MODULE 1

Sampling, Quantization and PCM

21 questions on sampling, quantization and pcm. Write your reasoning in the space under each one.

D1-01 · SAMPLING RATE, RESOLUTION AND BIT RATE

A message signal is bandlimited to $W = 12$ kHz.

- a) Give the Nyquist rate.
- b) The sampler is run 25% above the Nyquist rate. Give the rate and the sampling interval.
- c) Express the extra margin in part (b) as a guard band f_g .

D1-02 · SAMPLING RATE, RESOLUTION AND BIT RATE

An audio source is sampled at $f_s = 44$ kHz and each sample is quantized by a uniform quantizer with $L = 512$ levels.

- a) Give the number of bits per sample.
- b) Give the bit rate of the resulting PCM stream.
- c) How long does one bit last?

D1-03 · SAMPLING RATE, RESOLUTION AND BIT RATE

A signal $x(t)$ is bandlimited to 15 kHz. It is used to modulate a carrier:

$$y(t) = x(t) \cos(2\pi(60000)t).$$

- a) Sketch, in words, where the spectrum of y sits.
- b) Give the Nyquist rate of $y(t)$.

D1-04 · RECONSTRUCTION, ALIASING AND THE ANTI-ALIASING FILTER

A message bandlimited to $W = 15$ kHz is sampled at exactly the Nyquist rate and reconstructed with an ideal lowpass filter.

- a) Give the passband gain and cut-off of the reconstruction filter.
- b) Give its impulse response.
- c) Show that the reconstructed signal equals the message at every sampling instant.

D1-05 · RECONSTRUCTION, ALIASING AND THE ANTI-ALIASING FILTER

A pure tone at 7 kHz is sampled at $f_s = 10$ kHz without an anti-aliasing filter. The samples pass through an ideal lowpass filter with a 5 kHz cut-off.

- a) Give the frequencies of the replicas the sampler produces below 10 kHz.
- b) Give the frequency of the tone that comes out of the filter.
- c) What anti-aliasing filter would have prevented this, and what would have been lost?

D1-06 · A WAVEFORM THROUGH A UNIFORM QUANTIZER

The signal $m(t) = 3 \cos t$ is quantized by a uniform quantizer that spans its full range, with $R = 6$ bits per sample.

- a) Give the number of levels and the step size.
- b) Give the SQNR in decibels.

D1-07 · A WAVEFORM THROUGH A UNIFORM QUANTIZER

A full-scale sinusoid is to be quantized so that the SQNR is at least 40 dB.

- a) Give the smallest number of bits per sample that achieves it.
- b) Give the SQNR actually obtained.
- c) By how many decibels would the requirement be missed with one bit fewer?

D1-08 · A SOURCE DENSITY, A QUANTIZER, AND THE SQNR

The samples of a source are uniformly distributed on $[-2, 2]$ and are quantized by a uniform quantizer with $L = 64$ levels covering that range.

- a) Give the signal power and the step size.
- b) Give the quantization noise power.
- c) Give the SQNR in decibels.

D1-09 · A SOURCE DENSITY, A QUANTIZER, AND THE SQNR

The samples of a stationary source have the triangular density

$$f_X(x) = c \left(1 - \frac{|x|}{3} \right), \quad |x| \leq 3,$$

and zero elsewhere.

- a) Determine c .
- b) Give the power of the samples.

D1-10 · FROM A WAVEFORM TO A BIT STREAM

A non-negative signal is known to lie in $[0, 6]$ volts. It is quantized by an eight-level uniform quantizer covering that range, with the level of each tread at its midpoint, and encoded in natural binary.

- a) Give the step size and list the eight levels.
- b) Give the level and the code word for the samples 0.9, 2.4, 4.7 and 5.9 volts.

D1-11 · FROM A WAVEFORM TO A BIT STREAM

Sixteen quantization levels are to be encoded in four bits, either in natural binary or in a Gray code.

- a) Give the natural binary and Gray words for levels 7 and 8.
- b) Count how many bits change between those two adjacent levels under each code.
- c) Say what this costs when a channel error moves a decision to a neighbouring level.

D1-12 · A WAVEFORM THROUGH A UNIFORM QUANTIZER

A quantizer is designed for a peak amplitude $m_{\max}$, but the sinusoid actually presented to it reaches only a quarter of that.

- Give α in the two cases and the loss in decibels.
- How many extra bits per sample would recover the loss?

D1-13 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

A sinusoidal message is $x(t) = V_{\max} \cos(6000\pi t)$, where $V_{\max}$ is its peak amplitude. A uniform quantizer gives noise on $(-\frac{\Delta}{2}, \frac{\Delta}{2})$. The noise must not exceed $\pm 2\%$ of the peak-to-peak message amplitude. A 32-level PAM system encodes the quantized data. The sampling rate equals the Nyquist rate.

- Determine the minimum number of bits per sample.
- Calculate the bit rate of the system.
- Calculate the symbol rate of the system.

D1-14 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

The signal

$$x(t) = 6 \cos(2000\pi t) \cos(6000\pi t)$$

is sampled and quantized with a 512-level uniform quantizer. Assume $1 \text{ kbit/s} = 1000 \text{ bit/s}$.

- Determine the minimum sampling rate if a guard band of 2 kHz is required.
- Calculate the bit rate of the system at that sampling rate.
- If the bit rate is required to be 100 kbit/s, what should the guard band be?
- Calculate the step size of the uniform quantizer.

D1-15 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

A signal $X(t)$ has a bandwidth of 2.5 MHz. It is sampled, quantized and binary encoded to obtain a PCM signal.

- a) Determine the sampling rate if $X(t)$ is sampled at a rate 40% greater than the Nyquist rate.
- b) If the samples are quantized with a uniform quantizer with 4096 levels, determine the number of bits required per sample.
- c) Calculate the bit rate of this system in bits per second.
- d) How much of the bit rate is the margin above Nyquist responsible for?

D1-16 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

A stationary source is sampled, and the samples have the density

$$f_X(x) = k(1 + |x|), \quad x \in [-1, 1],$$

and zero elsewhere. The samples are quantized with a uniform quantizer with 128 levels covering $[-1, 1]$, and the quantization noise is uniform on $(-\frac{\Delta}{2}, \frac{\Delta}{2})$.

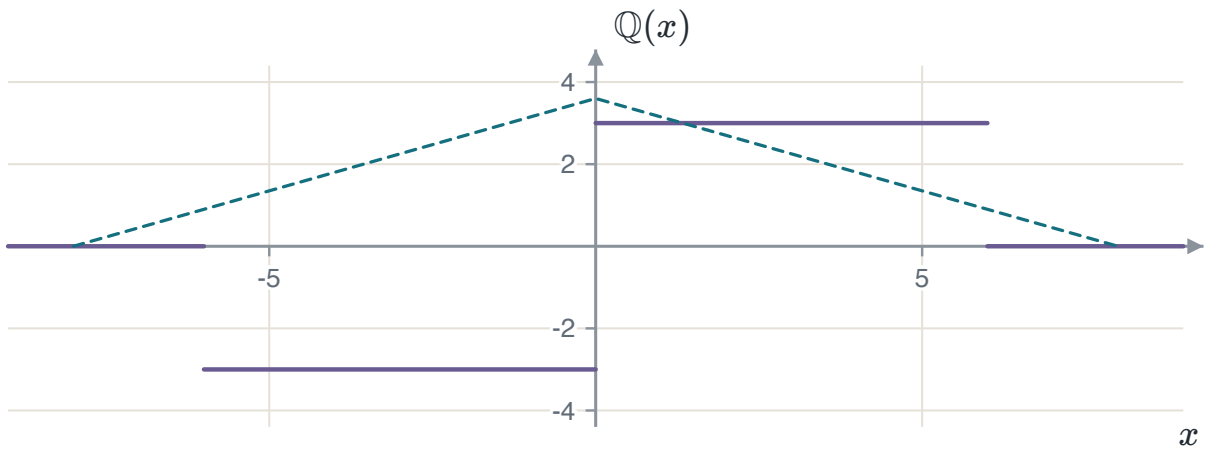
- a) Determine the value of k .
- b) Obtain the SQNR in decibels.
- c) For a 4 kHz source sampled at the Nyquist rate, calculate the bit rate of the corresponding PCM system.

The samples of a stationary source X have the triangular density

$$f_X(x) = c \left(1 - \frac{|x|}{8} \right), \quad |x| \leq 8,$$

and zero elsewhere. They are quantized by

$$\hat{X} = \mathbb{Q}(X) = \begin{cases} -3, & -6 < X < 0 \\ 3, & 0 < X < 6 \\ 0, & \text{otherwise.} \end{cases}$$



- Determine the value of c .
- Calculate the power of the samples of the source.
- Calculate the power of the quantization noise.
- Obtain the SQNR in decibels.

The signal

$$x(t) = 2 \cos(2000\pi t) + \cos(6000\pi t)$$

is sampled at the Nyquist rate and uniformly quantized with 128 levels.

- a) Calculate the bit rate of this system.
- b) Calculate the step size of the uniform quantizer.
- c) Calculate the SQNR of the quantization scheme in decibels.

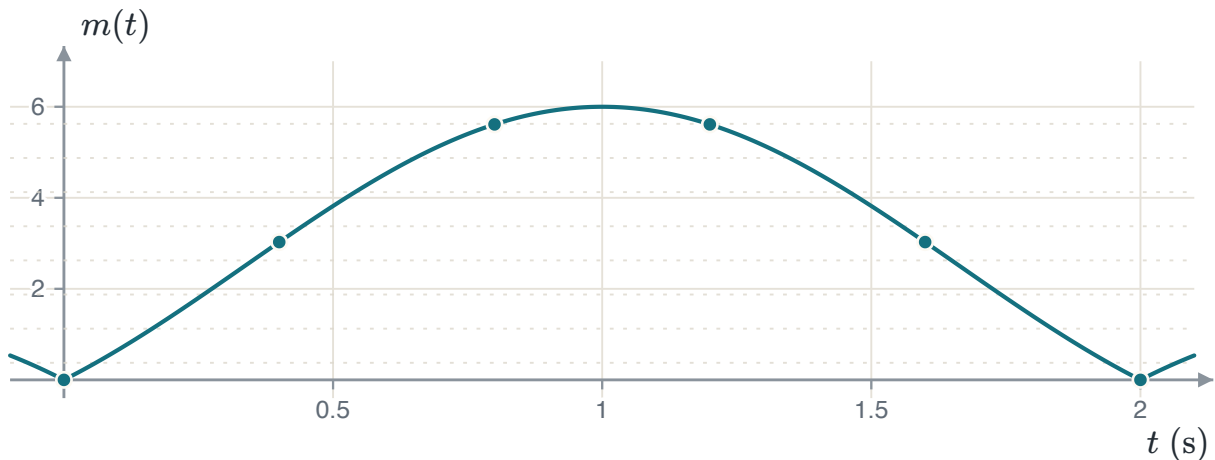
A message $x(t)$ is bandlimited to 20 kHz. The available converter runs at $f_s = 30$ kHz.

- a) Give the Nyquist rate, and say whether 30 kHz is adequate.
- b) A component of the message at 18 kHz is present. Where does it appear after sampling and reconstruction with an ideal lowpass filter of cut-off 15 kHz?
- c) Give the cut-off of an anti-aliasing filter that makes 30 kHz adequate, and say what is lost.
- d) With that anti-aliasing filter in place and a guard band of 4 kHz, give the sampling rate that would be needed.

The message is

$$m(t) = 6 |\text{sinc}(t - 1)|, \quad \text{sinc}(x) = \frac{\sin \pi x}{\pi x}.$$

It is sampled every $T_s = 0.4$ s. An eight-level uniform quantizer covers $[0, 6]$, with each level at the midpoint of its interval. The encoder uses natural binary coding.



- Give the step size and the eight levels.
- Give the six samples from $t = 0$ to $t = 2$ s, their levels and their code words.
- Give the bit rate and the duration of one bit.
- Sketch, in words, the polar NRZ waveform of the first two code words.

D1-21 · FROM A WAVEFORM TO A BIT STREAM

A 512×512 greyscale image is stored at 8 bits a pixel. It is then requantized more coarsely, and finally coded in pairs of neighbouring pixels.

- Give the stored size at 8 bits a pixel, in bits and in KiB.
- Give the size at $L = 32$ levels, the fraction saved, and what it costs in SQNR.
- At $L = 16$, neighbouring pixels never differ by more than one level. Give the number of pairs a coder must be able to name if it ignores that, and the number if it does not.
- Give the bits a pixel each of those two needs, and say what the saving was bought with.

MODULE 2

Baseband Transmission

20 questions on baseband transmission. Write your reasoning in the space under each one.

D2-01 · DESIGNING THE MATCHED FILTER

A rectangular pulse has amplitude $A = 2$ V and duration $T = 4$ ms. It is received in white Gaussian noise with two-sided density $N_0/2$, where $N_0 = 10^{-3}$ W/Hz.

- a) Give the impulse response of the matched filter.
- b) Give the filter output at $t = T$.
- c) Give the peak signal-to-noise ratio, in decibels.

D2-02 · DESIGNING THE MATCHED FILTER

A ramp pulse $s(t) = A t/T$ is used on $0 \leq t \leq T$ and zero elsewhere.

- a) Give its energy.
- b) Give the impulse response of the matched filter.
- c) By how many decibels does it fall short of a rectangular pulse of the same peak amplitude?

D2-03 · DESIGNING THE MATCHED FILTER

A half-sine pulse $s(t) = B \sin(\pi t/T)$ on $0 \leq t \leq T$ is to give the same matched-filter performance as a rectangular pulse of amplitude A and the same duration.

- a) Give the energy of each pulse.
- b) Give the peak amplitude B the half-sine needs.

D2-04 · ERROR PROBABILITY AGAINST ENERGY PER BIT

An antipodal baseband system has $E_b = 4 \times 10^{-6}$ J per bit and noise of two-sided density $N_0/2$ with $N_0 = 10^{-6}$ W/Hz. The two symbols are equally likely.

- a) Give E_b/N_0 in decibels.
- b) Give the bit error probability.

D2-05 · ERROR PROBABILITY AGAINST ENERGY PER BIT

An antipodal system is required to achieve a bit error probability of 10^{-5} .

- a) Give the argument of Q that produces it.
- b) Give the required E_b/N_0 , as a ratio and in decibels.

D2-06 · CONDITIONAL DENSITIES, THRESHOLD AND ERROR PROBABILITY

An antipodal system has $E_b = 1$ and $N_0 = 0.2$. The symbol s_0 is sent with probability 0.8.

- a) Give the optimal threshold.
- b) Say which way it has moved, and why that is the right direction.

D2-07 · ERROR PROBABILITY AGAINST ENERGY PER BIT

A system runs at $E_b/N_0 = 9$ dB. The bit rate is then raised by a factor of four with the transmit power and the noise density unchanged.

- a) Give the new E_b/N_0 in decibels.
- b) Give the bit error probability before and after.

D2-08 · A BINARY DECISION UNDER A NOISE DENSITY THAT IS NOT GAUSSIAN

A binary system has decision statistic $Y = s_m + N$, with $s_0 = -1$, $s_1 = +1$, equal priors, and noise of density

$$f_N(n) = \frac{1}{2}e^{-|n|}, \quad n \in \mathbb{R}.$$

The threshold is at zero.

- Give $P(\text{error} \mid s_0)$.
- Give the average error probability, and compare it with the Gaussian case of the same noise variance.

D2-09 · BANDWIDTH, THE NYQUIST CRITERION AND THE EYE

A baseband system transmits at $R_b = 20$ kbit/s.

- Give the Nyquist bandwidth.
- Give the transmission bandwidth with a raised-cosine roll-off of $\alpha = 0.35$.
- Give the excess bandwidth as a percentage.

D2-10 · BANDWIDTH, THE NYQUIST CRITERION AND THE EYE

A channel of bandwidth 24 kHz is to carry a raised-cosine baseband signal with roll-off $\alpha = 0.5$.

- Give the largest Nyquist bandwidth that fits.
- Give the largest bit rate.

D2-11 · BANDWIDTH, THE NYQUIST CRITERION AND THE EYE

A system uses the ideal Nyquist pulse $p(t) = \text{sinc}(2Wt)$ with $W = 1/(2T_b)$.

- Verify Nyquist's criterion in the time domain.
- Verify it in the frequency domain.

D2-12 · CONDITIONAL DENSITIES, THRESHOLD AND ERROR PROBABILITY

A correlator multiplies the received signal by a unit-energy basis function $\psi(t)$ and integrates over $0 \leq t \leq T_b$. The noise $w(t)$ is white and Gaussian with two-sided density $N_0/2$.

- a) Show that the noise contribution $n = \int_0^{T_b} w(\tau)\psi(\tau)d\tau$ has zero mean.
- b) Show that its variance is $N_0/2$.

D2-13 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

An additive white Gaussian noise channel has two-sided power spectral density 9 W/Hz. Two equiprobable messages use $s_0(t) = -3$ V and $s_1(t) = +3$ V for $0 \leq t \leq T_b = 2$ s. A matched-filter demodulator uses the unit-energy rectangular basis $\psi(t) = 1/\sqrt{T_b}$.

- a) Give the impulse response of the matched filter.
- b) Give the conditional probability density functions $f_Y(y | s_0)$ and $f_Y(y | s_1)$.
- c) Give the optimal decision threshold.
- d) Give the average probability of error.

D2-14 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

A binary system uses on-off signalling: $s_1(t) = A$ for $0 \leq t \leq T_b$ when "1" is sent, and $s_0(t) = 0$ when "0" is sent. The two are equally likely and the noise is white and Gaussian with two-sided density $N_0/2$.

- a) Give the two points in signal space and the average energy per bit.
- b) Give the optimal threshold.
- c) Give the bit error probability in terms of E_b/N_0 .
- d) Compare it with antipodal signalling at the same E_b/N_0 .

D2-15 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

In a binary system the decision statistic is $Y = s_m + N$ with $s_0 = -4$ and $s_1 = +4$. The noise has the triangular density

$$f_N(n) = c \left(1 - \frac{|n|}{10} \right), \quad |n| \leq 10,$$

and zero elsewhere. The two symbols are equally likely and the threshold is at zero.

- a) Determine c .
- b) Give $P(\text{error} \mid s_0)$.
- c) Give the average probability of error.
- d) Give the optimal threshold, with a reason.

D2-16 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

In a binary system, $Y = -2 + N$ for symbol "0" and $Y = +2 + N$ for symbol "1". The noise N is exponential with mean 4:

$$f_N(n) = \frac{1}{4}e^{-n/4}, \quad n \geq 0.$$

The density is zero for $n < 0$. The symbols are equally likely. The detector selects "1" when $Y > \lambda$.

- a) Give $P(\text{error} \mid s_1)$ and $P(\text{error} \mid s_0)$ as functions of λ , for $\lambda \leq 2$.
- b) Give the value of λ that minimises the average error probability.
- c) Give the average error probability there.
- d) Say what is different about this answer compared with the Gaussian case.

D2-17 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

In a binary pulse amplitude modulation system the correlator output is $y = \pm\sqrt{E_b} + n$, where n is Gaussian with zero mean and variance $N_0/2$. The symbol s_1 occurs with probability 0.25, and $E_b = 2$, $N_0 = 0.4$.

- a) Give the optimal threshold.
- b) Give the two conditional error probabilities.
- c) Give the average probability of error, as a number.
- d) Compare it with the error probability a threshold at zero would give.

D2-18 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

A detector observes $Y = N$ when "0" is sent and $Y = 4 + N$ when "1" is sent, where N is Gaussian with zero mean and variance 4. The symbol "0" occurs with probability 0.6.

- a) Give the optimal decision threshold.
- b) Give the two conditional error probabilities.
- c) Give the average probability of error.
- d) Give the average probability of error with the threshold placed midway, and say what the difference is worth.

D2-19 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

A baseband system must carry $R_b = 64$ kbit/s through a channel of bandwidth 48 kHz using raised-cosine pulse shaping.

- a) Give the Nyquist bandwidth.
- b) Give the largest roll-off factor that fits in the channel.
- c) Give the transmission bandwidth actually used at that roll-off, and the excess bandwidth.
- d) Say what would go wrong at $\alpha = 0$, which also fits.

D2-20 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

The overall pulse of a baseband system is normalised to $p(0) = 1$ and has been measured at the neighbouring sampling instants: $p(\pm T_b) = 0.10$ and $p(\pm 2T_b) = -0.04$, with everything beyond negligible. The noise at the detector has standard deviation $\sigma = 0.15$.

- a) Give the worst-case intersymbol interference at a sampling instant.
- b) Give the eye opening.
- c) Give the margin over noise, in standard deviations.
- d) Estimate the resulting error probability, and say what dominates it.

MODULE 3

Geometric Representation of Signals

20 questions on geometric representation of signals. Write your reasoning in the space under each one.

D3-01 · TESTING A SET FOR ORTHONORMALITY

Two waveforms are defined on $[0, 2]$: $p_1(t) = 1$ on $[0, 1]$ and zero after. $p_2(t) = 1$ on $[1, 2]$ and zero before.

- a) Are they orthogonal?
- b) Are they orthonormal?

D3-02 · TESTING A SET FOR ORTHONORMALITY

On $[0, T]$, take $\psi_1(t) = \sqrt{2/T} \cos(2\pi t/T)$ and $\psi_2(t) = \sqrt{2/T} \sin(2\pi t/T)$.

- a) Show that each has unit energy.
- b) Show that the two are orthogonal.

D3-03 · COORDINATES, ENERGY AND DISTANCE

With the basis $\psi_1 = 1/\sqrt{2}$ on $[0, 2]$ and $\psi_2 = 1$ on $[2, 3]$, a signal is $s(t) = 3$ on $[0, 2]$ and $s(t) = -2$ on $[2, 3]$.

- a) Give the coordinates of s .
- b) Give its energy, twice: from the vector and from the waveform.

D3-04 · COORDINATES, ENERGY AND DISTANCE

Two signals have coordinates $\mathbf{s}_1 = (3, 4)$ and $\mathbf{s}_2 = (-3, 4)$ against an orthonormal basis.

- a) Give the energy of each.
- b) Give the distance between them and the energy of their difference.

D3-05 · HOW MANY DIMENSIONS A SET NEEDS

Four waveforms are defined on $[0, 4]$: $s_1 = 1$ on $[0, 1]$; $s_2 = 1$ on $[1, 2]$; $s_3 = 1$ on $[2, 3]$; $s_4 = s_1 + s_2 + s_3$.

- a) How many dimensions does this set need?
- b) How many matched filters does a receiver for it need?

D3-06 · HOW MANY DIMENSIONS A SET NEEDS

Three waveforms on $[0, T]$ are $s_1(t) = A$, $s_2(t) = -2A$ and $s_3(t) = 3A$, each constant over the whole interval.

- a) How many dimensions does this set need?
- b) Give the basis and the three signal vectors.

D3-07 · READING A CONSTELLATION

A constellation has four points at $(\pm a, \pm a)$ against an orthonormal basis.

- a) Give the energy of each signal and the average energy.
- b) Give the smallest distance between any two points.
- c) Express that distance in terms of the average energy.

D3-08 · READING A CONSTELLATION

A constellation has three points at $(0, 1)$, $(\sqrt{3}/2, -1/2)$ and $(-\sqrt{3}/2, -1/2)$.

- a) Give the energy of each point.
- b) Give the distance between any two of them.

D3-09 · RUNNING GRAM-SCHMIDT

A signal set begins with $s_1(t) = 2$ on $[0, 4]$ and zero elsewhere.

- a) Give the first basis function.
- b) Give the first coordinate of s_1 .

D3-10 · RUNNING GRAM-SCHMIDT

With $\psi_1(t) = 0.5$ on $[0, 4]$ already found, the next signal is $s_2(t) = 3$ on $[0, 4]$ and zero elsewhere.

- a) Give the projection s_{21} .
- b) Give the remainder g_2 and say whether a second axis is added.

D3-11 · TESTING A SET FOR ORTHONORMALITY

On $[0, 2]$, take $\phi_1(t) = 1$ throughout, and $\phi_2(t) = 1$ on $[0, 1]$ with $\phi_2(t) = -1$ on $[1, 2]$.

- a) Are ϕ_1 and ϕ_2 orthogonal?
- b) Are they orthonormal? If not, give the orthonormal pair.

D3-12 · COORDINATES, ENERGY AND DISTANCE

Two signals have coordinates $\mathbf{x} = (1, 2, -1)$ and $\mathbf{y} = (3, 0, 1)$ against the same orthonormal basis.

- a) Give $\int x(t)y(t) dt$.
- b) Are the two signals orthogonal?

D3-13 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

Three waveforms are defined on $[0, 3]$: $s_1(t) = 2$ on $[0, 1]$. $s_2(t) = 2$ on $[1, 3]$. $s_3(t) = 2$ on $[0, 3]$. Each is zero outside the interval named.

- a) Use Gram–Schmidt to find an orthonormal basis.
- b) Give the three signal vectors.
- c) Draw the constellation and give the energy of each signal.
- d) Give the distance between the two closest signals.

D3-14 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

Three waveforms on $[0, 2]$: $s_1(t) = 1$ throughout. $s_2(t) = 1$ on $[0, 1]$ and -1 on $[1, 2]$. $s_3(t) = 2$ on $[0, 1]$ and 0 after.

- a) Find an orthonormal basis by Gram–Schmidt.
- b) Give the three signal vectors.
- c) Give the energy of each signal from its vector, and check one of them against the waveform.
- d) Say how many matched filters a receiver for this set needs.

D3-15 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

A four-point constellation has $\mathbf{s}_1 = (a, 0)$, $\mathbf{s}_2 = (0, a)$, $\mathbf{s}_3 = (-a, 0)$ and $\mathbf{s}_4 = (0, -a)$, with the four symbols equally likely.

- a) Give the energy of each signal and the average energy per symbol.
- b) Give the average energy per bit.
- c) Give the smallest distance between two points.
- d) Compare that distance with the one a square constellation of the same average energy gives.

D3-16 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

Two waveforms on $[0, T]$ are $s_1(t) = A \cos(2\pi t/T)$ and $s_2(t) = A \sin(2\pi t/T)$.

- Show that the two are orthogonal.
- Give an orthonormal basis for the set.
- Give the two signal vectors and their energies.
- Give the distance between them, and compare it with the distance an antipodal pair of the same energy would have.

D3-17 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

Against an orthonormal basis $\{\psi_1, \psi_2\}$, a signal set is $\mathbf{s}_1 = (2, 1)$, $\mathbf{s}_2 = (-1, 2)$ and $\mathbf{s}_3 = (1, -2)$, with the three symbols equally likely.

- Give the energy of each signal and the average.
- Give all three pairwise distances.
- Which pair is a receiver most likely to confuse, and why?
- Give the waveform $s_2(t)$ explicitly if $\psi_1(t) = 1/\sqrt{2}$ on $[0, 2]$ and $\psi_2(t) = 1$ on $[2, 3]$.

D3-18 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

A signal set consists of M waveforms $s_i(t) = A_i p(t)$, all multiples of the same finite-energy pulse $p(t)$ of energy E_p .

- How many dimensions does the set need, whatever M is?
- Give the basis function and the general signal vector.
- Give the energy of s_i .
- Give the distance between s_i and s_k .

D3-19 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

Four waveforms on $[0, 4]$ are built from the unit pulses $p_k(t) = 1$ on $[k-1, k]$, $k = 1, 2, 3, 4$:

$$s_1 = p_1 + p_2, \quad s_2 = p_3 + p_4, \quad s_3 = p_1 + p_3, \quad s_4 = p_2 + p_4.$$

- a) Are the four pulses p_k an orthonormal set? Give a reason.
- b) Give the four signal vectors against that basis.
- c) Give the energy of each signal and all the distinct pairwise distances.
- d) How many dimensions does the set $\{s_1, \dots, s_4\}$ actually need?

D3-20 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

Two engineers design different waveform sets. The first uses rectangular pulses on $[0, T]$. The second uses half-sine pulses on the same interval. Both sets turn out to have the constellation $\{(\sqrt{E}, 0), (0, \sqrt{E}), (-\sqrt{E}, 0), (0, -\sqrt{E})\}$.

- a) What is the same about the two systems?
- b) What is not?
- c) Give the average energy per symbol and the smallest distance.
- d) Which of the two would you choose, and on what grounds?

MODULE 4

The Optimal Receiver in AWGN

21 questions on the optimal receiver in awgn. Write your reasoning in the space under each one.

D4-01 · APPLYING THE DECISION RULE

A receiver with equally likely signals observes $\mathbf{r} = (0.8, -0.3)$. The signal points are $\mathbf{s}_1 = (1, 1)$, $\mathbf{s}_2 = (-1, 1)$, $\mathbf{s}_3 = (-1, -1)$ and $\mathbf{s}_4 = (1, -1)$.

- a) Give the squared distance to each point.
- b) Say which signal the receiver decides.

D4-02 · APPLYING THE DECISION RULE

A binary receiver has $\mathbf{s}_0 = -1$ and $\mathbf{s}_1 = +1$ on one axis, with $N_0 = 0.5$. The priors are $P(\mathbf{s}_0) = 0.8$ and $P(\mathbf{s}_1) = 0.2$. The observation is $r = 0.15$.

- a) Give the MAP metric for each signal.
- b) Say which signal the receiver decides, and what an equal-prior receiver would have decided.

D4-03 · A PAIRWISE ERROR PROBABILITY

Two signal points are 2.4 apart and the noise has two-sided density $N_0/2$ with $N_0 = 0.2$.

- a) Give the argument of Q for the pairwise error probability.
- b) Give the probability.

D4-04 · A PAIRWISE ERROR PROBABILITY

A binary antipodal system operates at $E_b/N_0 = 8$ dB.

- a) Give the distance between the two signal points in terms of E_b .
- b) Give the bit error probability.
- c) Give the answer if the same two waveforms were made orthogonal instead of antipodal, at the same E_b .

D4-05 · THE UNION BOUND IN ITS FOUR FORMS

A constellation has $M = 8$ points with $d_{\min} = 1.2$ and an average of $N_{\min} = 2$ points at that distance. The noise has $N_0 = 0.1$.

- a) Give the nearest-neighbour estimate of the symbol error probability.
- b) Give the minimum-distance bound.
- c) Say which to quote and why.

D4-06 · THE UNION BOUND IN ITS FOUR FORMS

Three equally likely signal points sit at $\mathbf{s}_1 = (0, 1)$, $\mathbf{s}_2 = (1, 0)$ and $\mathbf{s}_3 = (-1, 0)$, with $N_0 = 0.08$.

- a) Give the three pairwise distances.
- b) Give the general union bound on the symbol error probability.

D4-07 · COMPARING CONSTELLATIONS AT EQUAL ENERGY

Two constellations carry the same average symbol energy E_s . Four points on a square at $(\pm a, \pm a)$, and four points on a circle of radius ρ at 90° intervals.

- a) Give a and ρ in terms of E_s .
- b) Give $d_{\min}$ for each.
- c) Which performs better?

D4-08 · COMPARING CONSTELLATIONS AT EQUAL ENERGY

Square 16-QAM places its points at $(\pm d/2, \pm 3d/2)$ and the other sign combinations, so the spacing between neighbours is d . Its average symbol energy works out to $E_s = 2.5 d^2$.

- a) Give $d_{\min}$ in terms of E_s .
- b) Give the average number of nearest neighbours.
- c) Give the nearest-neighbour estimate of P_e at $E_s/N_0 = 15$ dB.

D4-09 · DRAWING THE DECISION REGIONS

Three equally likely signal points lie on a line at -2 , 0 and $+2$.

- a) Give the decision regions.
- b) Say which symbol is most often mistaken, and why.

D4-10 · DRAWING THE DECISION REGIONS

Two signal points sit at ± 1 on a line with $N_0 = 0.4$. The priors are $P(s_0) = 0.7$ for the point at -1 and $P(s_1) = 0.3$.

- a) Give the position of the boundary.
- b) Say which region has grown, and by what fraction of the separation.

D4-11 · APPLYING THE DECISION RULE

Three signals have energies $E_1 = 4$, $E_2 = 9$ and $E_3 = 1$, and the correlations with the observation are $\mathbf{r} \cdot \mathbf{s}_1 = 3.0$, $\mathbf{r} \cdot \mathbf{s}_2 = 5.0$ and $\mathbf{r} \cdot \mathbf{s}_3 = 2.0$. The signals are equally likely.

- a) Give the correlation metric for each.
- b) Say which signal the receiver decides, and which it would have chosen on correlation alone.

D4-12 · COMPARING CONSTELLATIONS AT EQUAL ENERGY

M -ary phase-shift keying places M points evenly on a circle of radius $\sqrt{E_s}$.

- Give $d_{\min}$ in terms of M and E_s .
- Give the nearest-neighbour estimate of P_e .
- By how many decibels does the required E_s/N_0 grow when M goes from 4 to 8 at a fixed error rate?

D4-13 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

Four equally likely signals are transmitted over an additive white Gaussian noise channel with $S_n(f) = N_0/2$. Their signal vectors are $\mathbf{s}_1 = (2, 0)$, $\mathbf{s}_2 = (0, 2)$, $\mathbf{s}_3 = (-2, 0)$ and $\mathbf{s}_4 = (0, -2)$.

- Draw the constellation and the optimal decision regions.
- Give the average symbol energy and the energy per bit.
- Give the minimum distance and the average number of nearest neighbours.
- Give the nearest-neighbour approximation to the symbol error probability as a function of E_b/N_0 .

D4-14 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

An M-ary scheme uses the four equally likely signal vectors $\mathbf{s}_1 = (1, 1)$, $\mathbf{s}_2 = (1, -1)$, $\mathbf{s}_3 = (-1, 1)$ and $\mathbf{s}_4 = (-3, -3)$, with $N_0 = 0.25$.

- Give the average symbol energy.
- Give all six pairwise distances and identify $d_{\min}$.
- Give the general union bound on the symbol error probability.
- Say which symbol is best protected and why.

D4-15 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

A binary system has $\mathbf{s}_0 = -\sqrt{E_b}$ and $\mathbf{s}_1 = +\sqrt{E_b}$ with $E_b = 1$ and $N_0 = 0.25$. The symbol $\mathbf{s}_0$ is sent with probability 0.75.

- Give the optimal decision boundary.
- Give the two conditional error probabilities.
- Give the average error probability.
- Compare it with the equal-prior receiver applied to the same source.

D4-16 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

Eight-point phase-shift keying places its symbols on a circle of radius $\sqrt{E_s}$ at 45° intervals, and each symbol carries three bits.

- Give the minimum distance in terms of E_s .
- Give the nearest-neighbour estimate of the symbol error probability.
- Evaluate it at $E_s/N_0 = 13$ dB.
- Give the bit error probability if Gray coding is used, and say what that assumes.

D4-17 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

A receiver observes $\mathbf{r}$ and must decide among M equally likely signals in additive white Gaussian noise.

- Write the exact expression for the average probability of a correct decision.
- Say why it cannot usually be evaluated.
- Give the union bound and say why it is an upper bound rather than an equality.
- Say when the bound stops being useful.

D4-18 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

Four-level pulse amplitude modulation places its symbols at $\pm d/2$ and $\pm 3d/2$ on one axis, and its average symbol energy is $E_s = 1.25 d^2$.

- Give the decision regions.
- Give the average number of nearest neighbours.
- Give the nearest-neighbour estimate of the symbol error probability in terms of E_s/N_0 .
- Evaluate it at $E_s/N_0 = 12$ dB, and compare with QPSK at the same E_s/N_0 .

D4-19 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

A constellation has six equally likely points at the corners of a regular hexagon of circumradius ρ , with N_0 given.

- Give the average symbol energy and the minimum distance.
- Give the average number of nearest neighbours.
- Give the nearest-neighbour estimate and the minimum-distance bound.
- By what factor do the two differ, and why?

D4-20 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES ABOVE

Two designers must carry two bits per symbol at an average symbol energy E_s . One proposes QPSK: four points on a circle of radius $\sqrt{E_s}$. The other proposes four points on a line, at $\pm a$ and $\pm 3a$.

- Give a in terms of E_s .
- Give $d_{\min}$ for each scheme.
- Give the ratio of their error probabilities at high signal-to-noise ratio.
- Say what the one-dimensional scheme gets in exchange.

D4-21 • THE UNION BOUND IN ITS FOUR FORMS

Three equally likely signals sit on one basis function at 0 , d and $2.2d$. The noise is white and Gaussian with two-sided density $N_0/2$, and $d^2/2N_0 = 9$.

- a) Give the decision regions, and name the neighbours that give each region a face.
- b) Write the intelligent union bound on the symbol error probability.
- c) Evaluate it, and evaluate the nearest-neighbour approximation beside it.
- d) Say which of the two a designer should quote, and why.

MODULE 5

Digital Modulation Methods

20 questions on digital modulation methods. Write your reasoning in the space under each one.

D5-01 · A BINARY SCHEME

A binary phase-shift keying system runs at $E_b/N_0 = 9$ dB.

- a) Give the two constellation points and the distance between them.
- b) Give the bit error probability.

D5-02 · A BINARY SCHEME

A binary frequency-shift keying system uses two orthogonal waveforms at the same $E_b/N_0 = 9$ dB.

- a) Give the two constellation points and the distance between them.
- b) Give the bit error probability.
- c) Give the penalty in decibels against BPSK.

D5-03 · A BINARY SCHEME

A binary amplitude-shift keying system sends nothing for a 0 and $\sqrt{2E/T_b} \cos(2\pi f_c t)$ for a 1, with the two bits equally likely. The system runs at $E_b/N_0 = 10$ dB, where E_b is the *average* energy per bit.

- a) Give the average energy per bit in terms of E .
- b) Give the distance between the two points in terms of E_b .
- c) Give the bit error probability.

D5-04 · A BINARY SCHEME

A link must achieve $P_b = 10^{-5}$.

- a) Give the E_b/N_0 in decibels that BPSK needs.
- b) Give the E_b/N_0 in decibels that BFSK needs.

D5-05 · M-ARY PHASE-SHIFT KEYING

An 8-PSK system operates at $E_s/N_0 = 13$ dB.

- a) Give $d_{\min}$ in terms of E_s .
- b) Give $N_{\min}$.
- c) Give the symbol error probability.

D5-06 · M-ARY PHASE-SHIFT KEYING

A QPSK system operates at $E_s/N_0 = 12$ dB.

- a) Give the symbol error probability.
- b) Give the corresponding E_b/N_0 in decibels.
- c) Give the bit error probability of BPSK at that same E_b/N_0 , and comment.

D5-07 · M-ARY PHASE-SHIFT KEYING

A design moves from QPSK to 8-PSK, holding the symbol error probability fixed.

- a) Give the ratio of the E_s/N_0 the two schemes need.
- b) Give the cost in decibels, per symbol and per bit.

D5-08 · M-ARY PHASE-SHIFT KEYING

Compare 8-PSK and 16-PSK at the same E_s .

- a) Give $d_{\min}$ for each in terms of $\sqrt{E_s}$.
- b) Give the ratio of their squared distances in decibels.

D5-09 · M-ARY AMPLITUDE-SHIFT KEYING

A 4-PAM system has points at $\pm A$ and $\pm 3A$, and runs at $E_s/N_0 = 12$ dB.

- a) Give the average symbol energy in terms of A .
- b) Give $N_{\min}$.
- c) Give the symbol error probability.

D5-10 · M-ARY AMPLITUDE-SHIFT KEYING

Compare 4-PAM and 8-PAM at the same average symbol energy.

- a) Give the ratio of their squared minimum distances.
- b) Give the cost of the move in decibels.

D5-11 · M-ARY AMPLITUDE-SHIFT KEYING

An 8-PAM constellation is used.

- a) Give $N_{\min}$.
- b) Say which symbols contribute least to the error probability, and why.

D5-12 · QUADRATURE AMPLITUDE MODULATION

A 16-QAM system operates at $E_s/N_0 = 15$ dB.

- a) Give $d_{\min}$ in terms of $E_{s,\text{avg}}$.
- b) Give $N_{\min}$.
- c) Give the symbol error probability.

D5-13 · QUADRATURE AMPLITUDE MODULATION

Compare 16-QAM with 16-PAM at the same average symbol energy. Both carry four bits a symbol.

- a) Give the squared minimum distance of each in terms of E_s .
- b) Give the advantage of QAM in decibels.

D5-14 · QUADRATURE AMPLITUDE MODULATION

A 64-QAM constellation is an eight-by-eight square grid.

- a) Give the number of corner, edge and interior points.
- b) Give $N_{\min}$.

D5-15 · COMPARING TWO SCHEMES

An 8-FSK system uses eight orthogonal waveforms at $E_s/N_0 = 10$ dB.

- a) Give the distance between any two constellation points.
- b) Give $N_{\min}$.
- c) Give an estimate of the symbol error probability.

D5-16 · COMPARING TWO SCHEMES

Compare 16-QAM and 16-FSK on how many bits per second each carries in one hertz of bandwidth. Take the bandwidth of a QAM signal as $1/T_s$ and that of M -FSK as $M/(2T_s)$.

- a) Give the bits per second per hertz of 16-QAM.
- b) Give the same for 16-FSK.
- c) Say what each scheme pays for what it gets.

D5-17 • COMPARING TWO SCHEMES

A link must achieve a symbol error probability of 10^{-4} .

- a) Give the E_b/N_0 in decibels that QPSK needs.
- b) Give the E_b/N_0 in decibels that 16-QAM needs.
- c) Say what the extra energy buys.

D5-18 • A FULL-LENGTH QUESTION ON ONE SCHEME

A system uses 16-QAM at a symbol rate of 10^6 symbols per second with $E_s/N_0 = 18$ dB.

- a) Give the bit rate.
- b) Give $d_{\min}^2$ in terms of E_s , and $N_{\min}$.
- c) Give the symbol error probability.
- d) Give the approximate bit error probability, assuming Gray coding.

D5-19 • A FULL-LENGTH QUESTION ON ONE SCHEME

A designer must send three bits a symbol and is choosing between 8-PSK and 8-PAM at the same average symbol energy.

- a) Give $d_{\min}^2$ for each in terms of E_s .
- b) Give $N_{\min}$ for each.
- c) Give the symbol error probability of each at $E_s/N_0 = 15$ dB.
- d) Say which to choose and why.

A channel allows $E_b/N_0 = 10$ dB and the link must reach a symbol error probability of 10^{-3} .

- a) Say whether QPSK meets the requirement.
- b) Say whether 8-PSK meets it.
- c) Say whether 16-QAM meets it.
- d) Choose the scheme that carries the most bits and still meets it.

MODULE 6

An Introduction to Information Theory

26 questions on an introduction to information theory. Write your reasoning in the space under each one.

D6-01 · THE INFORMATION IN ONE SYMBOL

A source emits a symbol of probability $1/4$ and, independently, a second of probability $1/32$.

- a) Give the information each symbol carries.
- b) Give the information in the pair, two ways.

D6-02 · THE INFORMATION IN ONE SYMBOL

A symbol carries 3 bits of self-information.

- a) Give its probability.
- b) Say what happens to the information if the probability is halved.

D6-03 · THE ENTROPY OF A SOURCE

A discrete memoryless source has probabilities 0.5, 0.25, 0.125, 0.125.

- a) Give the entropy.
- b) Compare it with the maximum an alphabet of this size allows.

D6-04 · THE ENTROPY OF A SOURCE

A source has probabilities 0.6, 0.3, 0.1.

- a) Give the entropy.
- b) Say how many bits a symbol would cost if the three symbols were simply numbered.

D6-05 · THE ENTROPY OF A SOURCE

A binary source emits 0 with probability 0.9 and 1 with probability 0.1.

- a) Give the entropy.
- b) Say what the entropy would be at $p = 0.5$, and comment on the shape of the curve.

D6-06 · THE ENTROPY OF A SOURCE

A source has eight symbols.

- a) Give the largest entropy it can have, and the probabilities that achieve it.
- b) If the measured entropy is 2.5 bits a symbol, give the efficiency of a fixed-length code that numbers the eight symbols.

D6-07 · AN EXTENDED SOURCE

The source with probabilities 0.7, 0.2, 0.1 is extended by three.

- a) Give the number of symbols in S^3 .
- b) Give the entropy of S^3 .
- c) Give the entropy per original symbol.

D6-08 · AN EXTENDED SOURCE

A source has four equally likely symbols and is extended by two.

- a) Give $H(S)$ and $H(S^2)$.
- b) Give the probability of each symbol of S^2 , and verify the entropy directly.

D6-09 · MEASURING A CODE

A source with probabilities 0.5, 0.25, 0.125, 0.125 is coded as 0, 10, 110, 111.

- a) Give the average codeword length.
- b) Give the efficiency.
- c) Say whether the code is a prefix code.

D6-10 · MEASURING A CODE

A four-symbol source is coded as $s_1: 0$, $s_2: 10$, $s_3: 110$, $s_4: 111$.

- a) Decode the string 01101110.
- b) Say why no backtracking was needed.

D6-11 · THE KRAFT INEQUALITY

Three sets of codeword lengths are proposed for a four-symbol source: $\{1, 2, 3, 3\}$, $\{2, 2, 2, 2\}$ and $\{1, 2, 2, 3\}$.

- a) Apply the Kraft inequality to each.
- b) Say which admit a prefix code.

D6-12 · THE KRAFT INEQUALITY

A prefix code for five symbols must have $l_1 = 1$, $l_2 = 2$ and $l_3 = 3$.

- a) Give the Kraft budget already spent.
- b) Give the shortest possible equal lengths for l_4 and l_5 .

D6-13 · MEASURING A CODE

A source has entropy $H(S) = 2.35$ bits a symbol. Three codes are claimed for it, with average lengths 2.20, 2.35 and 3.40 bits.

- a) Say which claims are possible.
- b) Say which of the possible ones a prefix code could have produced.

D6-14 · AN EXTENDED SOURCE

A coder must reach within 0.05 bits a symbol of the entropy by coding blocks.

- a) Give the smallest block length that guarantees it.
- b) Give the size of the codebook if the source has three symbols.

D6-15 · BUILDING AND JUDGING A HUFFMAN CODE

Build a Huffman code for a source with probabilities 0.5, 0.25, 0.125, 0.125.

- a) Give the merges.
- b) Give the codeword lengths and the average length.
- c) Give the efficiency.

D6-16 · BUILDING AND JUDGING A HUFFMAN CODE

Build a Huffman code for a source with probabilities 0.4, 0.3, 0.2, 0.1.

- a) Give the merges and the codeword lengths.
- b) Give the average length and the entropy.
- c) Give the efficiency.

D6-17 · BUILDING AND JUDGING A HUFFMAN CODE

Two Huffman codes for the same source both have $\bar{L} = 2.2$ bits. The first has lengths 2, 2, 2, 3, 3 with probabilities 0.4, 0.2, 0.2, 0.1, 0.1. The second has lengths 1, 2, 3, 4, 4 for the same probabilities in the same order.

- a) Give the variance of each.
- b) Say which is preferred and why.

D6-18 · BUILDING AND JUDGING A HUFFMAN CODE

A discrete memoryless source has probabilities 0.4, 0.2, 0.2, 0.1, 0.1.

- a) Give the entropy.
- b) Build the minimum-variance Huffman code and give the lengths.
- c) Give the average length and the efficiency.
- d) Give the variance, and compare with a fixed-length code.

D6-19 · BUILDING AND JUDGING A HUFFMAN CODE

A source has six symbols with probabilities 0.3, 0.25, 0.2, 0.12, 0.08, 0.05.

- a) Give the entropy.
- b) Build a Huffman code and give the lengths.
- c) Give the average length, the efficiency and the variance.
- d) Compare with a fixed-length code.

D6-20 · BUILDING AND JUDGING A HUFFMAN CODE

A binary source emits 0 with probability 0.8 and 1 with probability 0.2.

- a) Give the entropy.
- b) Give the average length and efficiency of a Huffman code on single symbols.
- c) Extend the source by two and give the average length per original symbol of a Huffman code on the pairs.
- d) Say what the extension bought.

D6-21 · PARSING FOR A UNIVERSAL CODE

A Lempel–Ziv encoder begins with 0 at position 1 and 1 at position 2 in its dictionary. Parsing a stream produces, in order, the new pieces 00, 01, 011 and 10. Pointers are sent in three bits.

- a) Give the dictionary position of each new piece.
- b) Give the pointer and the innovation bit for each.
- c) Write the four transmitted blocks.
- d) Decode the block 0111 and say what position the result takes.

D6-22 · A CHANNEL AND WHAT IT LETS THROUGH

A binary channel has $p(y_0 | x_0) = 0.9$ and $p(y_0 | x_1) = 0.2$. The transmitter sends x_0 with probability 0.6.

- a) Write the channel matrix and check it.
- b) Give the four joint probabilities.
- c) Give the output distribution.
- d) Give $H(Y | X)$, and say whether it would change if the transmitter changed.

D6-23 · A CHANNEL AND WHAT IT LETS THROUGH

A binary symmetric channel has crossover probability $p = 0.2$ and equally likely inputs.

- a) Give $H(X)$ and $H(Y | X)$.
- b) Give $H(Y)$.
- c) Give the mutual information.
- d) Say what fraction of each transmitted bit the channel destroys.

D6-24 · CAPACITY, AND WHAT IT PERMITS

A binary symmetric channel with crossover probability 0.05 is used 10^6 times a second.

- a) Give the capacity in bits per channel use.
- b) Give it in bits per second.
- c) Say whether 800 kbit/s can be sent with an arbitrarily small error probability.
- d) Give the largest crossover probability for which 800 kbit/s would be possible.

D6-25 · CAPACITY, AND WHAT IT PERMITS

A coherent binary PSK link runs at $E_b/N_0 = 6$ dB. The receiver makes a hard decision on every bit and hands the result to a decoder.

- a) Give the crossover probability of the binary symmetric channel the decoder sees.
- b) Give the capacity of that channel in bits per channel use.
- c) The link carries 1 Mbit/s of channel bits. Give the largest information rate that can be sent reliably.
- d) Say what the hard decision has thrown away.

A channel has bandwidth 3.4 kHz and a signal-to-noise ratio of 30 dB.

- a) Give its capacity.
- b) Give the capacity if the transmitted power is doubled, and the gain as a percentage.
- c) Give the capacity if the bandwidth is doubled instead, at the same noise density, and the gain as a percentage.
- d) Say which is the better buy, and why the two answers differ so much.